

Multiple Choice (10 marks)

1. The effects of iron on certain aspects of socio-emotional development (shyness, orientation/engagement and response to unfamiliar pictures) is due to:
 - (a) The role of iron in serotonin production
 - (b) None of the above
 - (c) The role of iron in neurochemistry
 - (d) The role of iron in maintaining blood sugar levels
 - (e) The role of iron in muscle function
2. Consanguinity shows a strong association with which pattern of inheritance?
 - (a) Codominant inheritance
 - (b) Mitochondrial inheritance
 - (c) Y-linked recessive
 - (d) X-linked dominant
 - (e) Autosomal dominant
3. Which products of ADP degradation increase in concentration in the blood during multiple sprint sports?
 - (a) Ammonia, urea and creatine.
 - (b) Ammonia, hypoxanthine and creatinine.
 - (c) Ammonia, creatine and uric acid.
 - (d) Urea, creatine and uric acid.
 - (e) Ammonia, urea and uric acid.
4. Which of the following structures should be visible on a dental panoramic radiograph?
 - (a) Only the mandibular canal and hard palate
 - (b) The mandibular canal, maxillary sinus, hard palate, zygomatic arch, and ethmoid sinus
 - (c) The mandibular canal, maxillary sinus, hard palate, zygomatic arch, and frontal sinus
 - (d) The mandibular canal and maxillary sinus
 - (e) The mandibular canal
5. A patient suffers a broken neck with damage to the spinal cord at the level of the sixth cervical vertebra.

- (a) It is impossible to predict an effect on breathing.
- (b) Their breathing rate will significantly increase.
- (c) They will only be able to breathe quietly.
- (d) Their breathing will be erratic and unpredictable.
- (e) They will be unable to breathe without life support.

True / False (10 marks)

Answer True or False

- 6. True or False: Motor neuron cell bodies are primarily located in Lamina V of the spinal cord.
- 7. True or False: Acyclovir is an antiviral medication specifically used to inhibit herpes viruses.
- 8. True or False: Impulses for pain travel primarily through the dorsal columns of the spinal cord.
- 9. True or False: If survival time with a disease increases, the prevalence of that disease in the population will also increase.
- 10. True or False: Saliva contains enzymes that primarily act on proteins, facilitating their digestion before they enter the stomach.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the process of fat digestion and absorption, focusing on the role of emulsification and the necessity of triglyceride breakdown for fatty acid transport.
- 12. Discuss the demographic trends of the global ageing population, focusing on the gender distribution in older age groups and its implications for health care systems.

End of Paper